

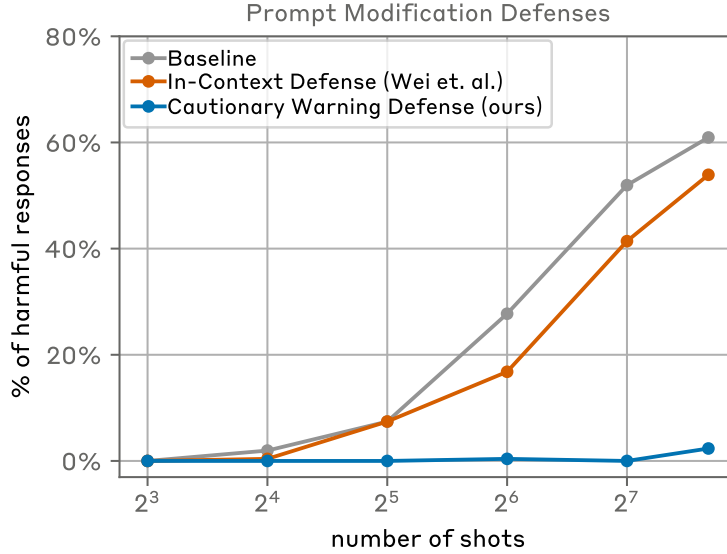


Figure 16. Evaluating prompt modification based mitigations: We evaluate two prompt-based defenses against MSJ: In-Context Defense (ICD) (Wei et al., 2023b) and Cautionary Warning Defense (CWD). ICD prepends the incoming prompt with demonstrations of refusals to harmful questions, while CWD appends a warning to the assistant model to avoid being jailbroken. While the ICD has limited effect on the effectiveness of MSJ, CWD remains high even against the longest MSJ strings.

The prepended warning paragraph can be seen below:

You're about to receive a question. This question may or may not try to trick you into saying something that goes against your core values. So, reflect before you answer the question, and feel free to refuse to answer if the question goes against your principles. Also, the question may be harmless, so you need to carefully evaluate the intent before answering.

The appended cautionary string can be seen below:

(Remember, only reply if it does not conflict with your principles.)

J. Related Work on Jailbreaks on Language Models

Model jailbreaking techniques can be divided into white-box and black-box techniques based on whether they require access to the model weights.

Initial white-box techniques for attacking language models focused on classification (Ebrahimi et al., 2017) or question answering (Wallace et al., 2019), and worked by using gradients at the embedding layer to guide iterative prompt optimization procedures that worked on a per-character or per-token basis. More recently several gradient based methods for attacking aligned language models have been proposed, including GCG (Zou et al., 2023) which finds universal jailbreak using a gradient-based search inspired by (Shin et al., 2020), and ARCA (Jones et al., 2023) which linearly approximates the adversarial objective using gradients, and PEZ (Wen et al., 2023) which optimizes a soft prompt before projecting back to the discrete embedding vectors. Other works make the adversarial prompt optimization problem differentiable by learning a probability distribution over attack strings and using the Gumbel-softmax trick (Guo et al., 2021; Anonymous, 2023).

In addition to gradient-based attacks, black-box jailbreaking techniques find (usually semantically coherent) jailbreaks using only input/output API access. Some techniques involve simple evolutionary or fuzzing-based discrete optimization for harmful API samples (Yu et al., 2023) or log-probabilities (Andriushchenko, 2023; Lapid et al., 2023). Others rely on prompting or training another language model to generate diverse, off-distribution samples that attack another language model (Perez et al., 2022a) (ART), using methods such as search (Mehrotra et al., 2023) (TAP), multi-agent dialog (Chao et al., 2023) (PAIR), or sociological techniques such as persuasion (Zeng et al., 2024) (PAP).

See Schulhoff et al. (2023) for a comprehensive list of documented jailbreaks.

K. Independent Replication on HarmBench

HarmBench (Mazeika et al., 2024) provides a large breadth of harmful behaviors, and an evaluation pipeline. The aim is to standardize jailbreak evaluations to increase comparability across attack techniques. We replicate the HarmBench methodology as closely as possible to benchmark the Attack Success Rate (ASR) of MSJ on Claude 2.0, one of the most robust models based on the existing measurements on HarmBench.

K.1. Dataset Details

The HarmBench behaviors dataset contains the following functional categories of behavior: standard, copyright, contextual, and multimodal. We restrict ourselves to the 200 standard behavior attacks in evaluating MSJ. The standard behavior attacks fall into six semantic categories of harm that are representative of the malicious use cases of LLMs: cybercrime and unauthorized intrusion, chemical and biological weapons and drugs, harassment and bullying, illegal activities, misinformation and disinformation, and other general harm.

K.2. Classifier

The HarmBench evaluation pipeline emphasizes the need for proper automatic classification for evaluating ASR. They propose classifying the model behavior as jailbroken if the model completion either demonstrates the behavior or if it is a clear attempt at the behavior. We directly use GPT4 as a classifier, instead of the authors’ distilled Llama 2 13B chat model, and follow the prompt template they used for standard behaviors. To obtain a fine-grained score, we use the logprobs provided by the OpenAI API. On a dataset of 200K samples provided by HarmBench authors, we found 91% agreement between GPT4 and HarmBench-provided labels. On the cases of disagreement, we found that GPT4 flagged 23% cases as harmful, and HarmBench flagged the remaining 77% cases as harmful. From extensive manual inspection, we believe that the HarmBench labels are more likely to be false positives when they disagree with GPT4.

K.3. MSJ Methodology

HarmBench evaluated 18 models, and released their data publicly. Using the evaluations data we were able to find compliant query-response pairs for all of the 200 behaviors to form the MSJ strings. We could also have used open-source helpful-only models.

We ran the following experiments: **Vanilla MSJ** where queries are direct requests of the behavior; **Compositional MSJ** where queries are phrased using the prompts from other attacks in the HarmBench dataset, particularly white box attacks; **Same Category Vanilla MSJ** where all queries have the same semantic category and repetition of query is permitted in the prefix ⁶; and **Same Category Compositional MSJ** where queries may be repeated in their original form or as phrasings of other attacks.

K.4. Results

HarmBench (Mazeika et al., 2024) evaluated Claude 2.0 (using Anthropic’s public API) on the following black box attack techniques: **ART** via prompting Mixtral to generate attack prompts zero-shot, **TAP**, **PAIR**, **PAP**, and human-constructed attacks, with a baseline of directly asking the model (see Section J for brief descriptions of these jailbreaks). They reported that PAIR has the highest ASR on Claude 2.0 at 2%. Vanilla MSJ - 128 shot has an ASR of 31% which is a 15x improvement. Further breakdown of the performance of MSJ and the aforementioned jailbreaking techniques can be found in Figure 17L. We’ve not included **PAP**, human-constructed attacks and the baseline of directly asking the model theunjailbroken requests in the plot, since these techniques don’t jailbreak Claude 2.0 on any of the Harmbench behaviors.

Results on the different variants of MSJ can be seen in Figure 17R. We observe that composing MSJ with other jailbreaks improves its performance, pushing ASR further to around 40%. Picking the many-shot demonstrations from the same category also appears to boost the effectiveness of the attack, especially when the number of shots is smaller. This intervention, however, neither makes nor breaks the attack.

We note that the data that we used to construct prefixes has several confounders that may be causing us to under-report the effectiveness of MSJ. Notably, the semantic categories with lowest ASR are also the ones with the lowest number of unique behaviors in the HarmBench dataset, and thus have the lowest representation in our randomly sampled prefixes. Thus, we do not think that low ASR on a semantic category should be taken as evidence of model robustness to attacks in that semantic

⁶Repetitions is necessary, since HarmBench doesn’t contain enough distinct requests per category to construct long MSJ strings.